

Termwerte berechnen Klasse 7 · Lösungen

1b) 6; 11; 46

a) $5 \cdot 2 - 4 = 10 - 4 = 6$; b) $5 \cdot 3 - 4 = 15 - 4 = 11$;
c) $5 \cdot 10 - 4 = 50 - 4 = 46$

2. 13; 20; 11; 8

a) $7 + 2 \cdot 3 = 7 + 6 = 13$; b) $4 \cdot (3 + 2) = 4 \cdot 5 = 20$;
c) $20 - 3 \cdot 3 = 20 - 9 = 11$; d)
 $2 \cdot 3 + 3 - 1 = 6 + 3 - 1 = 8$

3. 1; 4; 7; 10; 13

$3 \cdot 1 - 2 = 1$; $3 \cdot 2 - 2 = 4$; $3 \cdot 3 - 2 = 7$;
 $3 \cdot 4 - 2 = 10$; $3 \cdot 5 - 2 = 13$

x	1	2	3	4	5
$3x - 2$	1	4	7	10	13

4. nein, 22 €; es fehlen 2 €

$4 + 3 \cdot 6 = 4 + 18 = 22$ €. Das Geld reicht nicht, es
fehlen $22 - 20 = 2$ €.

5b) 4; -5; 7

a) $3 \cdot (-2) + 10 = -6 + 10 = 4$; b)
 $3 \cdot (-5) + 10 = -15 + 10 = -5$; c)
 $3 \cdot (-1) + 10 = -3 + 10 = 7$

6. 11; 6; 17; -9

a) $8 - (-3) = 8 + 3 = 11$; b) $-2 \cdot (-3) = 6$; c)
 $5 - 4 \cdot (-3) = 5 - (-12) = 5 + 12 = 17$; d)
 $-3 - 6 = -9$

7. 2; 18

a) $6 \cdot (-2 - 1) + 20 = 6 \cdot (-3) + 20 = -18 + 20 = 2$;
b) $10 - 2 \cdot (-7 + 3) = 10 - 2 \cdot (-4) = 10 - (-8) = 18$

8. $3 - 4x$ (Wert 11), um 3 größer

$3 - 4 \cdot (-2) = 3 + 8 = 11$; $2 \cdot (-2 + 6) = 2 \cdot 4 = 8$.
 $3 - 4x$ hat den größeren Wert, um $11 - 8 = 3$.

9b) 25; 25; 1; 100

a) $5^2 = 25$; b) $(-5)^2 = (-5) \cdot (-5) = 25$; c)
 $(-1)^2 = 1$; d) $(-10)^2 = 100$

10. 16; 8; 6

a) $4 \cdot (-2)^2 = 4 \cdot 4 = 16$; b) $(-2)^2 + 4 = 4 + 4 = 8$;
c) $(-2)^2 - (-2) = 4 + 2 = 6$

11. 1; -3

a) $10 - (-3)^2 = 10 - 9 = 1$; b)
 $2 \cdot (-1)^2 + 5 \cdot (-1) = 2 \cdot 1 + (-5) = 2 - 5 = -3$

12. 8; 3; 0; -1; 0; 3; negativ nur
für $x = 1$

$(-2)^2 - 2 \cdot (-2) = 4 + 4 = 8$;
 $(-1)^2 - 2 \cdot (-1) = 1 + 2 = 3$; $0^2 - 2 \cdot 0 = 0$;
 $1^2 - 2 \cdot 1 = 1 - 2 = -1$; $2^2 - 2 \cdot 2 = 0$;
 $3^2 - 2 \cdot 3 = 9 - 6 = 3$. Negativ nur für $x = 1$.

x	-2	-1	0	1	2	3
$x^2 - 2x$	8	3	0	-1	0	3

13b) 1; 11; -12; -7

a) $4 + (-3) = 1$; b) $2 \cdot 4 - (-3) = 8 + 3 = 11$; c)
 $4 \cdot (-3) = -12$; d) $-3 - 4 = -7$

14. -2

$\frac{1 - (-7)}{-4} = 8 : (-4) = -2$

15. 2

$\frac{-1 - 7}{-4} = (-8) : (-4) = 2$

16. $b - a$ (Wert 6)

$$4 - (-2) = 6. \text{ Die anderen: } a \cdot b = (-2) \cdot 4 = -8; \\ \frac{b}{a} = 4 : (-2) = -2; a - b = -2 - 4 = -6.$$

17. 14

$$6 - 2 \cdot (-4) = 6 - (-8) = 6 + 8 = 14$$

18. -20

$$4 \cdot (-3 - 2) = 4 \cdot (-5) = -20$$

19. 15

$$2 \cdot (-3)^2 + (-3) = 2 \cdot 9 - 3 = 18 - 3 = 15$$

20. -2

$$\frac{6 + (-2)}{-2} = 4 : (-2) = -2$$

21. 14

$$(-3)^2 - (-5) = 9 + 5 = 14$$

22. a) 18 m, ja b) 40 m, nein

a) $\frac{30}{10} = 3$; $3 \cdot 3 + 3^2 = 9 + 9 = 18$ m; $18 < 20$: ja, es
hält 2 m vor dem Kind. b) $\frac{50}{10} = 5$;
 $3 \cdot 5 + 5^2 = 15 + 25 = 40$ m; $40 > 20$: nein, der Weg
ist 20 m zu lang.